

State of the Art and Technical References

Social navigation of agricultural robots in shared corridors

343 valid runs / 9 methods / 4 scenarios

Purpose. Organize the conceptual, technological, and methodological foundations required to interpret social navigation, experimental evaluation, statistical inference, TOPSIS analysis, and Monte Carlo sensitivity. The document stands alone as theoretical and bibliographic support for the calculations.

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1 Scope and organization of evidence

Mobile-robot navigation in shared agricultural spaces requires geometric planning, reactive avoidance, social reasoning, operational continuity, and quantitative evaluation. A method may avoid collisions while still producing socially intrusive paths, excessive stopping, or deadlock in front of stationary workers. Evaluation must therefore consider safety, proxemics, efficiency, smoothness, and mission success together [4, 5, 6].

The evidence is organized into five complementary levels:

1. **Human foundation:** proxemic zones, body orientation, relative speed, and legibility.
2. **Algorithmic foundation:** global planning, local control, and avoidance families.
3. **Experimental foundation:** repeatable scenarios, run-level telemetry, and success criteria.
4. **Statistical foundation:** descriptives, robust tests, effect sizes, and categorical outcomes.
5. **Decision foundation:** multicriteria integration and sensitivity analysis.

Table 1: Evidence map and analytical function.

Axis	Technical question	Required evidence
Spatial safety	Does the robot maintain sufficient separation?	Minimum distance, zone occupancy, and scenario distribution.
Continuity	Does it complete the mission without deadlock?	Successes, failures, waiting time, and stops.
Efficiency	Does motion avoid unnecessary detours?	Path length, duration, and path efficiency.
Smoothness	Does control avoid oscillation?	Speed variability and numerical acceleration.
Decision	How does preference change with priorities?	TOPSIS and weight perturbation.

The bibliographic selection prioritizes foundational work for each family, surveys of social navigation, quantitative evaluation methods, and applications in agricultural robotics and shared environments. References explain hypotheses, assumptions, and limitations; they do not replace experimental data.

2 Proxemics and representation of human space

2.1 Interpersonal zones

Proxemics describes how people regulate interpersonal space. The classical division distinguishes intimate, personal, social, and public zones [1]. In robot navigation, these limits are operational guides rather than universal boundaries because they depend on culture, task, speed, orientation, visibility, and the possibility of evasion [2, 3].

Table 2: Proxemic zones used as operational references.

Zone	Lower limit	Upper limit	Navigation interpretation
Intimate	0.00 m	0.45 m	Close-contact intrusion requiring priority attention.
Personal	0.45 m	1.20 m	Sensitive region for encounters and passing maneuvers.
Social	1.20 m	3.60 m	Functional interaction with moderate separation.
Public	3.60 m	—	Distant interaction with low proxemic pressure.

Minimum distance d^* detects the most critical approach, whereas cumulative times $T_{personal}$ and $T_{intimate}$ quantify exposure duration. Mean distance does not replace these metrics: a path may have a high average and still contain a brief critical approach.

2.2 Anisotropy, orientation, and relative motion

An isotropic field assigns equal cost in every direction. This simplification ignores that a frontal approach may be perceived differently from lateral or rear passage. Anisotropic fields represent this asymmetry through longitudinal and lateral scales rotated with human orientation [18, 19].

A Gaussian representation can be written as

$$G(\mathbf{x}) = \exp \left[-\frac{1}{2} \left(\frac{x_h^2}{\sigma_{\parallel}^2} + \frac{y_h^2}{\sigma_{\perp}^2} \right) \right], \quad (1)$$

where (x_h, y_h) are coordinates in the human frame. If $\sigma_{\parallel} > \sigma_{\perp}$, the field penalizes the longitudinal direction over a longer range. Superposition can represent several people, but saturation, opposing gradients, and local minima must be controlled.

2.3 Implications for agricultural corridors

Vineyard corridors combine limited width, partial visibility, and workers who may not react to the robot. Reciprocity cannot be assumed automatically. Evaluation should therefore separate: (i) physical distance, (ii) time in sensitive zones, (iii) continuity of motion, and (iv) path efficiency. An overly conservative policy can maintain large separation and lose operational viability; a fast policy can preserve mobility while entering the intimate zone.

3 Global planning and local control

3.1 Geometric planning

Graph-based methods such as Dijkstra and A* compute routes over discrete representations [8, 9]. Sampling-based methods such as PRM and RRT explore more complex configuration spaces [10, 11, 12]. In traversable polygonal environments, a navigation mesh provides an efficient global route and separates environmental connectivity from local social decisions.

The global route does not solve human interaction by itself. Its role is to guide the robot toward the goal; the local controller converts that guidance into commands compatible with obstacles,

dynamics, and social costs.

3.2 Reactive families

Table 3: Conceptual comparison of local-control families.

Family	Principle	Strength	Risk or assumption
Potential fields	Attractive and repulsive gradients	Low cost and interpretable formulation	Local minima and oscillation.
DWA	Sampling of dynamically reachable velocities	Reactive control with explicit constraints	Dependence on weights and local horizon.
TEB	Optimization of a space-time band	Smooth paths and explicit constraints	Optimization cost and topological sensitivity.
VO/ORCA	Velocities outside collision cones	Efficient multiagent coordination	Reciprocity assumption.
Social force	Virtual goal and interaction forces	Interpretable collective behavior	Calibration and possible oscillation.
CBF	Safety constraints on control	Formal guarantee under model assumptions	Conservatism or loss of feasibility.

The dynamic-window approach evaluates reachable velocities over a short horizon [13]. TEB optimizes positions and timing along a deformable band [14]. Velocity obstacles exclude commands that would produce collision; ORCA distributes avoidance responsibility among agents [15, 16]. This distribution can degrade when a person remains stationary and does not share the maneuver.

4 Social navigation and represented families

4.1 Social force

The social-force model combines a goal-directed force with repulsion from people and obstacles [17]. Its principal advantages are interpretability and gradual separation. Behavior depends on gains, distance scales, damping, and speed saturation.

4.2 Control barriers

A barrier function $h(\mathbf{x})$ defines the safe set $\mathcal{C} = \{\mathbf{x} : h(\mathbf{x}) \geq 0\}$. A common condition requires

$$\dot{h}(\mathbf{x}, \mathbf{u}) + \alpha(h(\mathbf{x})) \geq 0, \quad (2)$$

and selects a control close to the nominal command subject to this inequality [20]. Barriers provide a formal separation between objective and safety, but a hard constraint may leave few admissible commands in congested corridors.

4.3 Discrete supervisors and continuous fields

Stop supervisors apply binary rules: advance or stop. They are simple but can deadlock in front of a stationary person when the release condition is never satisfied. Hysteresis reduces rapid switching

but does not remove structural deadlock. Continuous fields preserve nonzero commands and permit passing, provided that gradients, dynamic limits, and geometry yield a feasible direction.

Table 4: Families represented by the nine methods.

Code	Family	Main characteristic
M0	Geometric navigation	Global guidance without an explicit social layer.
M1	Threshold supervisor	Binary stop upon entering the activation region.
M2	Hysteresis supervisor	Separate entry and release thresholds.
M3	Continuous proxemic field	Continuous modulation without every anisotropic component.
M4	Complete anisotropic field	Orientation, multiagent aggregation, and minimum escape speed.
B1	Social DWA	Command sampling with social costs.
B2	ORCA/RVO	Reciprocal avoidance in velocity space.
B3	Social-force model	Interaction through virtual forces.
B4	Barrier-based control	Explicit safety constraint.

5 Agricultural robotics and operational safety

Autonomous agricultural robotics integrates localization, perception, planning, and control on irregular surfaces, narrow rows, and variable illumination and vegetation [22, 23, 24]. In the presence of people, physical safety must be accompanied by understandable behavior, mission continuity, and controlled stopping.

Three distinctions are essential:

- **Collision avoidance** is not equivalent to respecting personal space.
- **Large separation** does not guarantee continuity if the robot remains stopped.
- **Mission success** alone does not demonstrate smoothness or comfort.

In shared corridors, useful evaluation should include frontal encounters, lateral intrusions, social following, and multiagent congestion. These situations activate different assumptions: passing ability, response to a stationary worker, reciprocal coordination, and accumulation of social fields.

6 Digital environment, instrumentation, and metrics

An instrumented digital environment allows geometry, routes, speeds, human states, and termination criteria to be fixed. Controlled repetition supports comparisons under the same mission and produces homogeneous telemetry. The Unity–MATLAB architecture separates discrete-time execution from statistical processing: Unity produces states and events, while MATLAB consolidates scalars, tables, and tests.

Table 5: Recorded metric families.

Family	Metrics	Question
Proxemics	d^* , mean distance, low percentile, $T_{personal}$, $T_{intimate}$	How close, and for how long?
Mission	$T_{mission}$, path length L , efficiency	What spatial and temporal cost does the policy impose?
Kinematics	Mean speed, maximum speed, σ_v , acceleration	Is motion smooth and controllable?
Stopping	T_{stop} and N_{stop}	Are there persistent pauses or oscillations?
Outcome	Success/failure and reason	Is the mission completed within the limit?

The mission is defined as $A \rightarrow B \rightarrow A$ within 180 s. Continuous metrics are summarized only for successful runs, preventing a timeout from being interpreted as a completed trajectory. Success rate uses all valid runs.

7 Experimental reference design

The corpus contains 343 valid runs, nine methods, and four scenarios. Methods M0–M4 and baseline B1 provide available aggregates; B2, B3, and B4 have run-level scalars. This difference in availability determines which tests can be calculated without reconstructing observations.

Table 6: Evaluation scenarios.

Code	Scenario	Examined capability
E1	Frontal encounter	Anticipatory modulation and longitudinal separation.
E2	Lateral intrusion	Anisotropic response and passing ability.
E3	Social following	Stability behind a moving person.
E4	Multiagent congestion	Continuity with stationary workers and overlapping fields.

Balancing scenarios prevents a single pooled mean from concealing critical behavior. Minimum distance should be inspected by scenario when a family depends on reciprocity or geometry changes the available escape direction.

8 Statistical foundation

8.1 Descriptives and data availability

For each method and metric, sample size n , mean $\bar{x}$, and sample standard deviation s are retained. Descriptions must report denominators and success conditioning. Aggregated statistics support parametric comparisons of means; rank tests require individual observations.

8.2 Welch and Hedges

The nine-group comparison uses Welch ANOVA, reducing dependence on the equal-variance assumption [25]. For pairwise comparisons, Welch’s test uses

$$t_W = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}. \quad (3)$$

Hedges g corrects finite-sample bias in the standardized mean difference [26]. Its sign must be interpreted with criterion direction: larger values favor separation and success but penalize time, length, variability, and stopping criteria.

8.3 Multiplicity control

Benjamini–Hochberg orders $p_{(i)}$ values and retains through the largest index satisfying

$$p_{(i)} \leq \frac{i}{m}q, \quad (4)$$

thereby controlling the expected false-discovery rate [27]. Comparisons anchored on M4 use eight tests per metric.

8.4 Rank tests and ordinal effects

Kruskal–Wallis compares groups through ranks [28]. It is limited to B2/B3/B4, for which run-level scalars are available. Mann–Whitney U compares pairs of these groups [29]; Cliff $\delta = P(X > Y) - P(X < Y)$ expresses ordinal dominance from -1 to $+1$ [30].

8.5 Categorical outcomes

Fisher’s exact test evaluates 2×2 tables with small counts. Global method-by-outcome association is evaluated with

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad V = \sqrt{\frac{\chi^2}{N \min(r-1, c-1)}}. \quad (5)$$

Cramér V separates association magnitude from statistical significance. Rates are calculated from integer counts before rounding.

Table 7: Mapping between question and statistical procedure.

Question	Procedure	Data requirement
Does any mean differ among nine methods?	Welch ANOVA	n , $\bar{x}$, and s .
Which pairs differ relative to M4?	Welch, Hedges g , Benjamini–Hochberg	Group summary statistics.
Do B2/B3/B4 ranks differ?	Kruskal–Wallis	Run-level scalars.
How separated are two distributions?	Mann–Whitney and Cliff δ	Run-level scalars.
Do success probabilities differ?	Fisher exact	Success and failure counts.
Is there a global association?	Chi-square and Cramér V	Method-by-outcome table.

9 TOPSIS and Monte Carlo sensitivity

TOPSIS represents each method as a row of $X \in \mathbb{R}^{9 \times 7}$. The criterion vector is

$$(d^*, T_{personal}, T_{mission}, N_{stop}, \sigma_v, a_{max}^{num}, \eta_s).$$

d^* and η_s are benefits; all remaining criteria are costs [32].

The three weight vectors are

$$w_{W1} = (0.20, 0.15, 0.10, 0.15, 0.15, 0.10, 0.15), \quad (6)$$

$$w_{W2} = (0.22, 0.20, 0.05, 0.10, 0.20, 0.15, 0.08), \quad (7)$$

$$w_{W3} = (0.20, 0.15, 0.05, 0.10, 0.25, 0.20, 0.05). \quad (8)$$

Vector normalization and closeness are

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\sum_k x_{kj}^2}}, \quad v_{ij} = w_j \tilde{x}_{ij}, \quad CC_i = \frac{D_i^-}{D_i^+ + D_i^-}. \quad (9)$$

Ranking depends on weights and is not a universal superiority statement. To quantify local stability, every weight is uniformly perturbed and renormalized:

$$w'_j = w_j(1 + \epsilon_j), \quad \epsilon_j \sim U(-\epsilon_{max}, +\epsilon_{max}), \quad \hat{w}_j = \frac{w'_j}{\sum_k w'_k}. \quad (10)$$

The analysis uses $\epsilon_{max} = 0.10$ and 0.20 , with 2000 samples per level. Ideals, distances, coefficients, and ranks are recalculated in every replicate. First-rank frequency is

$$f_i^{(1)} = \frac{1}{S} \sum_{s=1}^S \mathbf{1}[r_i^{(s)} = 1], \quad S = 2000. \quad (11)$$

10 Critical synthesis of the evidence

Table 8: Strengths and limits by family.

Family	Favorable evidence	Condition limiting interpretation
Discrete supervisors	Simple implementation and predictable stopping.	A stationary person can keep the stop condition active.
Anisotropic fields	Directional separation and continuous modulation.	Field scales, aggregation, and saturation require calibration.
Social DWA	Integrates dynamics and social costs into sampling.	Results depend on weights and command resolution.

Family	Favorable evidence	Condition limiting interpretation
ORCA/RVO	Efficient coordination among reactive agents.	Reciprocity can fail with nonreactive workers.
Social force	Gradual and interpretable interaction.	Poor gains can produce oscillation or detours.
CBF	Formal safety constraints.	A strict barrier can reduce feasibility and continuity.
TOPSIS	Traceable integration of heterogeneous criteria.	Ranking changes with priorities and scales.

The evidence supports a trade-off interpretation: spatial safety, continuity, efficiency, and smoothness are not necessarily maximized by the same policy. Separate metrics, effect sizes, and sensitivity analysis are therefore required in addition to global success rate.

11 Limitations and reproducibility

11.1 Limitations

- Simulation does not reproduce every sensory, mechanical, and human source of field variability.
- Workers may remain stationary; other human-response models would produce different interactions.
- Available aggregates for M0–M4/B1 prevent nine-group rank tests without complete scalars.
- Numerical pseudo-acceleration may depend on sampling frequency and telemetry source.
- TOPSIS encodes priorities through weights; ranks must be accompanied by sensitivity.
- The references cover representative families but do not exhaust every MPC, learning, or social-planning variant.

11.2 Reproducibility checklist

1. Preserve method and scenario codes.
2. Record sample sizes and exclusion reasons.
3. Separate success-conditioned continuous metrics from binary counts.
4. Retain full precision and round only for display.
5. Apply Benjamini–Hochberg within each family of eight comparisons.
6. Restrict rank tests to groups with available scalars.
7. Preserve the TOPSIS matrix, criterion directions, and weights.
8. Fix and record the Monte Carlo seed.
9. Verify units, decimal locale, and absence of nonfinite values.

12 Conclusions

Agricultural social navigation requires a hierarchical architecture: global planning for connectivity, local control for interaction, and an evaluation layer that separates safety, comfort, continuity, and efficiency. No single metric summarizes all these objectives.

The nine-method comparison covers discrete supervision, continuous fields, social DWA, reciprocal avoidance, social force, and control barriers. The statistical framework distinguishes means, ranks, effects, and success rates according to data availability. TOPSIS integrates seven criteria, while Monte Carlo quantifies sensitivity to changes in priority. Together, these elements provide conceptual and methodological traceability for the calculations in Documents 01–06.

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